Front Polynomials of Ranked Recursive Factorization Systems

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Abstract

A ranked recursive factorization system (RRFS) associates to every atom a finite rank-decreasing rooted tree and to every object a finite rooted forest. This paper studies a one-variable polynomial obtained directly from the complete node fronts of these forests. If A is a complete front, we record only the number of terminal vertices selected by A and define

$$\mathcal{F}_h(T) = \sum_{A \in \mathrm{CF}(F_h)} T^{\lambda(A)}.$$

The first result is structural: this geometric definition automatically satisfies the recursion

$$\mathcal{F}_\tau(T) = T, \qquad \mathcal{F}_\alpha(T) = 1 + \prod_\beta \mathcal{F}_\beta(T)^{B_{\beta,\alpha}}, \qquad \mathcal{F}_h(T) = \prod_\alpha \mathcal{F}_\alpha(T)^{v_\alpha(h)}.$$

Thus the recursive polynomials used for numerical Pratt trees are not an independent definition; they are precisely terminal-front generating polynomials.

We introduce three notions. An RRFS is *front-stable* if every atomic front polynomial is irreducible in $\mathbb{Z}[T]$; it is *front-faithful* if, in addition, distinct atoms give nonassociate polynomials; and it is *evaluation-compatible* with a multiplicative norm N if there exists $u > 1$ with $\mathcal{F}_\alpha(u) = N(\alpha)$ for every atom. For a front-faithful RRFS the front polynomials themselves form a polynomial RRFS with the same transition matrix and the same recursive Green operator. If evaluation compatibility also holds, the front lift preserves the associated Dirichlet zeta function.

The numerical Pratt RRFS is the main positive example. Its front polynomials are the polynomials $f_n(T)$, Jonathan Love's irreducibility theorem gives p prime if and only if f_p is irreducible, distinct primes give distinct f_p, and $f_n(2) = n$. Hence the natural-number RRFS is front-faithful and its front lift is zeta-preserving, recovering the Riemann zeta function. We then construct further front-stable RRFS, including chains and cyclotomic stars. In contrast, the polynomial RRFS over finite fields and the ideal-Pratt RRFS over Dedekind rings are generally front-splitting or non-faithful. Explicit examples over $\mathbb{F}_3[x]$ and $\mathbb{Z}[i]$ show a single original atom whose front polynomial factors in $\mathbb{Z}[T]$. These examples suggest a new problem: to classify recursive geometries whose terminal-front transformation preserves irreducibility, factorization, and zeta data.

Status and scope. Complete node fronts, the numerical Pratt polynomials f_n, their front interpretation, and the irreducibility theorem for f_p are established in earlier work; the proof of irreducibility follows the root-location strategy due to Jonathan Love. Unique factorization in polynomial rings and Dedekind domains is classical. The purpose of the present note is the abstract RRFS formulation: defining the one-variable terminal-front polynomial geometrically, deriving its recursion from complete fronts, and organizing the resulting notions of front stability, front faithfulness, front lifts, and zeta transfer. The finite-field and Dedekind computations below are exact examples, not asymptotic claims.

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1 Introduction

A ranked recursive factorization system starts with ordinary unique factorization and then places a finite rank-decreasing recursive tree below each atom. Thus every object has two compatible descriptions: its top-level factorization into atoms and its unfolded recursive forest. Complete node fronts are maximal antichains in this forest. They cut every root-to-leaf path exactly once and therefore provide a natural family of combinatorial sections through the recursion.

For the numerical Pratt forest of a positive integer n, earlier work introduced polynomials $f_n(T)$ recursively by

$$f_1(T) = 1, \qquad f_2(T) = T,$$
$$f_p(T) = 1 + \prod_{q|p-1} f_q(T)^{v_q(p-1)} \quad (p > 2 \text{ prime}),$$

and

$$f_n(T) = \prod_p f_p(T)^{v_p(n)}.$$

The same polynomials have a geometric interpretation: $f_n(T)$ is the generating polynomial of complete node fronts, where the exponent of T counts how many terminal 2-vertices occur in the selected front. In particular, the recursive definition and the node-front definition coincide.

This equivalence suggests a general construction. Given any RRFS, define the front polynomial first from complete node fronts and only afterwards prove the recursive formula. This order is conceptually important. The recursion is not an additional axiom imposed on a polynomial family; it is forced by the root-versus-descendants decomposition of complete fronts.

Once this construction is available, an algebraic question appears naturally:

$$\boxed{\text{When is the front polynomial of every atomic tree irreducible?}}$$

For the natural-number Pratt system the answer is positive. This is precisely what makes prime detection by f_n possible. But the phenomenon is not universal. In finite-field polynomial RRFS and in ideal-Pratt systems over number rings, an atomic recursive tree can have a reducible front polynomial. Moreover, distinct atoms can have the same front polynomial because the one-variable statistic deliberately forgets root labels.

The present paper develops a vocabulary for these possibilities. The main chain is

$$\boxed{\begin{array}{c}\text{RRFS forest} \longrightarrow \text{complete node fronts} \longrightarrow \mathcal{F}_h(T) \\ \longrightarrow \text{front recursion} \longrightarrow \text{irreducibility / factorization / zeta transfer}\end{array}}$$

The examples show that the natural-number Pratt system is unusually rigid: its front transformation preserves both the atomic factorization and the natural norm.

2 RRFS background

Definition 2.1 (Ranked recursive factorization system)**.** *A ranked recursive factorization system is a quadruple*

$$\mathfrak{R} = (H, \mathcal{A}, D, \rho)$$

where:

(i) H is a reduced factorial commutative cancellative monoid with atom set $\mathcal{A}$;

(ii) $D : \mathcal{A} \to H$ is a predecessor map;

(iii) $\rho : \mathcal{A} \to \mathbb{N}_0$ is a rank;

(iv) if $\beta \mid D\alpha$, then $\rho(\beta) < \rho(\alpha)$.

An atom τ is terminal if $D\tau = 1$. The terminal set is denoted $\mathcal{T}$.

Every object $h \in H$ has a unique finite factorization

$$h = \prod_{\alpha \in \mathcal{A}} \alpha^{v_\alpha(h)}.$$

Write

$$D\alpha = \prod_{\beta \in \mathcal{A}} \beta^{B_{\beta,\alpha}}, \qquad B_{\beta,\alpha} = v_\beta(D\alpha).$$

Rank descent makes the directed dependency relation acyclic.

2.1 Atomic trees and object forests

For an atom α, its atomic tree T_α has a root labelled α and, for each β, exactly $B_{\beta,\alpha}$ children labelled β, with a copy of T_β below each child. Rank descent implies that every such tree is finite.

For

$$h = \prod_{\alpha} \alpha^{v_\alpha(h)},$$

the object forest is the disjoint union

$$F_h = \bigsqcup_{\alpha \in \mathcal{A}} v_\alpha(h) T_\alpha.$$

The total number of β-labelled vertices in F_h is the recursive coordinate $M_\beta(h)$. If B is the transition matrix, then

$$M(h) = (I - B)^{-1} v(h), \qquad v(h) = (I - B) M(h).$$

These identities are the boundary and recursive-coordinate relations of the RRFS.

3 Complete node fronts and the terminal-front polynomial

3.1 Complete fronts

Give every rooted tree its ancestor order: $u \preceq v$ if u lies on the root-to-v path. Vertices in different components of a forest are incomparable.

Definition 3.1 (Node front)**.** *A node front in a finite rooted forest F is an antichain for the ancestor order. It is* complete *if it meets every root-to-leaf path. The set of complete fronts is denoted* $\mathrm{CF}(F)$*.*

Since a root-to-leaf path is totally ordered, a node front meets a given path at most once. Thus a complete front meets every root-to-leaf path exactly once. Equivalently, complete fronts are maximal antichains in a finite rooted forest.

3.2 The one-variable terminal statistic

The polynomial studied here deliberately forgets all nonterminal labels. For a complete front $A \in \mathrm{CF}(F_h)$ define

$$\lambda(A) = \#\{v \in A : \ell(v) \in \mathcal{T}\},$$

the number of selected terminal vertices.

Definition 3.2 (Terminal-front polynomial)**.** *The front polynomial of $h \in H$ is*

$$\boxed{\mathcal{F}_h(T) = \sum_{A \in \mathrm{CF}(F_h)} T^{\lambda(A)} \in \mathbb{Z}[T].}$$

For an atom α write $\mathcal{F}_\alpha = \mathcal{F}_{T_\alpha}$.

This is the primary definition in this paper. The recursive formulas below are consequences of it.

Remark 3.3 (Several terminal types)**.** *If one wants to remember the different terminal labels, one can use the multivariate terminal polynomial*

$$\mathcal{F}_h(\{T_\tau\}_{\tau\in\mathcal{T}}) = \sum_{A \in \mathrm{CF}(F_h)} \prod_{\tau \in \mathcal{T}} T_\tau^{N_\tau(A)},$$

where $N_\tau(A)$ counts selected τ-vertices. The one-variable polynomial is the diagonal specialization $T_\tau = T$ for every terminal type. The present note focuses on this compressed one-variable statistic because irreducibility after compression is nontrivial.

3.3 The front recursion theorem

Theorem 3.4 (Geometric front recursion)**.** *Let $\mathfrak{R}$ be an RRFS.*

(i) If τ is terminal, then

$$\boxed{\mathcal{F}_\tau(T) = T.}$$

(ii) If α is nonterminal, then

$$\boxed{\mathcal{F}_\alpha(T) = 1 + \prod_{\beta \in \mathcal{A}} \mathcal{F}_\beta(T)^{B_{\beta,\alpha}}.}$$

(iii) For every object h,

$$\boxed{\mathcal{F}_h(T) = \prod_{\alpha \in \mathcal{A}} \mathcal{F}_\alpha(T)^{v_\alpha(h)}.}$$

Proof. If τ is terminal, T_τ consists of one vertex. Its unique complete front is the singleton root, which is terminal, so the polynomial is T.

Now let α be nonterminal. A complete front of T_α has exactly two forms. It may contain the root. Then no descendant can be selected and, since the root is nonterminal, this front contributes $T^0 = 1$. Otherwise the front omits the root. To meet every root-to-leaf path it must then choose a complete front independently in every child subtree. There are $B_{\beta,\alpha}$ copies of the child tree T_β, so the generating polynomials multiply and give the displayed product.

Finally, a complete front of a disjoint union of rooted forests is exactly a choice of one complete front in each connected component. Terminal counts add across components, so the generating polynomials multiply. Applying this to the $v_\alpha(h)$ copies of T_α gives the last identity. □

This theorem is the abstract reason why the numerical Pratt polynomials admit both a recursive definition and a node-front interpretation.

3.4 Immediate enumerative consequences

Proposition 3.5. *For every $h \in H$:*

(i) $\mathcal{F}_h(1) = \#\,\mathrm{CF}(F_h)$*;*

(ii) $\deg \mathcal{F}_h = \sum_{\tau \in \mathcal{T}} M_\tau(h)$*;*

(iii) $\mathcal{F}_h$ *is monic.*

Proof. At $T = 1$ every complete front has weight one. The set of all leaves is a complete front, and every leaf is terminal. Thus its exponent is the total terminal population $\sum_\tau M_\tau(h)$. Any front selecting a nonterminal node excludes all terminal descendants below that node and therefore has smaller terminal count. Hence the all-leaf front is the unique front of maximal degree, giving monicity. □

4 Front stability, faithfulness, and polynomial lifts

Theorem 3.4 turns the geometry of node fronts into a family of integer polynomials. We now ask whether ordinary factorization of these polynomials reflects the original atomic factorization of the RRFS.

Definition 4.1 (Front-stable, front-faithful, front-splitting)**.** *Let $\mathfrak{R}$ be an RRFS.*

- $\mathfrak{R}$ *is* front-stable *if $\mathcal{F}_\alpha(T)$ is irreducible in $\mathbb{Z}[T]$ for every atom α.*
- *It is* front-faithful *if it is front-stable and distinct atoms give nonassociate front polynomials.*
- *It is* front-splitting *if at least one atom α has reducible $\mathcal{F}_\alpha(T)$.*

Front stability is a property of the compressed terminal-front statistic, not of finiteness. Both finite and infinite RRFS can be stable or splitting.

Definition 4.2 (Front splitting defect)**.** *For an atom α, let*

$$\delta_{\mathfrak{R}}(\alpha) = \Omega_{\mathbb{Z}[T]}(\mathcal{F}_\alpha),$$

the number of irreducible factors of $\mathcal{F}_\alpha$ counted with multiplicity. Thus $\delta_{\mathfrak{R}}(\alpha) = 1$ exactly at front-stable atoms.

4.1 The front lift

Suppose $\mathfrak{R}$ is front-faithful. Let H_{fr} be the submonoid of $\mathbb{Z}[T] \setminus \{0\}$ generated by the irreducibles $\{\mathcal{F}_\alpha : \alpha \in \mathcal{A}\}$. Unique factorization in $\mathbb{Z}[T]$ makes this a free commutative monoid on those generators.

Theorem 4.3 (Front lift theorem)**.** *If $\mathfrak{R} = (H, \mathcal{A}, D, \rho)$ is front-faithful, then the map*

$$\Phi_{\mathrm{fr}} : H \longrightarrow H_{\mathrm{fr}}, \qquad h \longmapsto \mathcal{F}_h(T),$$

is a factorization-preserving monoid isomorphism onto its image. Moreover, H_{fr} carries a ranked recursive factorization system with

$$D_{\mathrm{fr}}(\mathcal{F}_\alpha) = \prod_\beta \mathcal{F}_\beta^{B_{\beta,\alpha}},$$

and the same rank ρ. For nonterminal α,

$$\boxed{\mathcal{F}_\alpha(T) = 1 + D_{\mathrm{fr}}(\mathcal{F}_\alpha)(T).}$$

The original and front-lift RRFS have the same transition matrix B, hence the same recursive Green operator $(I - B)^{-1}$.

Proof. By Theorem 3.4,

$$\mathcal{F}_h = \prod_\alpha \mathcal{F}_\alpha^{v_\alpha(h)}.$$

Front faithfulness and unique factorization in $\mathbb{Z}[T]$ imply that the exponent vector on the right uniquely determines $v(h)$, hence h. The displayed predecessor formula has factorization exponents $B_{\beta,\alpha}$ and therefore satisfies the same rank descent. The final identity is precisely the atomic front recursion. □

Remark 4.4. *If an RRFS is front-splitting, one may still factor each $\mathcal{F}_\alpha$ into irreducibles in $\mathbb{Z}[T]$. This produces a finer algebraic factorization, but it need not inherit a canonical predecessor map from the original RRFS. Thus "factor the front polynomial and declare the factors to be new RRFS atoms" is a new construction requiring additional work; it is not automatic.*

5 Front evaluation and zeta transfer

Every positive evaluation of the front polynomial gives a multiplicative gauge.

Definition 5.1 (Front evaluation gauge)**.** *Fix $u > 1$. Define*

$$N_{\mathrm{fr},u}(h) = \mathcal{F}_h(u).$$

By forest multiplicativity,

$$N_{\mathrm{fr},u}(h_1 h_2) = N_{\mathrm{fr},u}(h_1) N_{\mathrm{fr},u}(h_2).$$

Whenever it converges, define the front zeta function

$$\boxed{\zeta_{\mathrm{fr},\mathfrak{R}}(s;u) = \sum_{h\in H} \mathcal{F}_h(u)^{-s} = \prod_{\alpha\in\mathcal{A}} \left(1 - \mathcal{F}_\alpha(u)^{-s}\right)^{-1}.}$$

The Euler product uses the original atoms of H and therefore exists formally independently of front stability. Zeta *transfer* is stronger: it asks whether the natural norm of another RRFS is exactly reproduced by the front evaluation.

Definition 5.2 (Evaluation compatibility)**.** *Let $N : H \to \mathbb{R}_{>0}$ be a multiplicative norm. The front polynomial is evaluation-compatible with N if there exists $u > 1$ such that*

$$\boxed{\mathcal{F}_\alpha(u) = N(\alpha) \quad \text{for every atom } \alpha.}$$

Then automatically $\mathcal{F}_h(u) = N(h)$ for every h.

Theorem 5.3 (Zeta transfer)**.** *Let $(\mathfrak{R}, N)$ be a normed RRFS. If $\mathfrak{R}$ is front-faithful and evaluation-compatible at $u > 1$, then the front lift of Theorem 4.3 preserves the norm and, in every common half-plane of convergence,*

$$\boxed{\zeta_{\mathfrak{R},N}(s) = \sum_{h\in H} N(h)^{-s} = \sum_{P\in H_{\mathrm{fr}}} P(u)^{-s}.}$$

Equivalently, the Euler products correspond atom by atom.

Proof. Evaluation compatibility gives $\mathcal{F}_h(u) = N(h)$ by multiplicativity. Front faithfulness identifies h with $\mathcal{F}_h$ without merging or splitting atoms. The Dirichlet sums and Euler products therefore agree term by term under the front lift. □

6 Natural numbers: the front-faithful model

Let

$$H = \mathbb{N}, \qquad \mathcal{A} = \{2, 3, 5, \ldots\},$$

with

$$D(2) = 1, \qquad D(p) = p - 1 \quad (p > 2 \text{ prime}).$$

The unique terminal atom is 2. The atomic tree T_p is the usual Pratt tree of p, and the forest F_n contains $v_p(n)$ copies of T_p.

For a complete front A let $\lambda(A)$ be the number of selected 2-vertices. Define geometrically

$$\mathcal{F}_n(T) = \sum_{A \in \mathrm{CF}(F_n)} T^{\lambda(A)}.$$

Theorem 3.4 gives

$$\mathcal{F}_2(T) = T,$$
$$\mathcal{F}_p(T) = 1 + \prod_{q \mid p-1} \mathcal{F}_q(T)^{v_q(p-1)},$$

and

$$\mathcal{F}_n(T) = \prod_p \mathcal{F}_p(T)^{v_p(n)}.$$

These are exactly the defining recursions of the Pratt polynomial family $f_n(T)$. Hence

$$\boxed{\mathcal{F}_n(T) = f_n(T).}$$

This equality is the abstract model for the present paper: the recursive formula is a theorem forced by the complete-front geometry.

6.1 Irreducibility and prime detection

A theorem proved using the root-location argument due to Jonathan Love states that for every prime p, the polynomial $f_p(T)$ is irreducible in $\mathbb{Z}[T]$. Consequently

$$\boxed{n \text{ is prime} \iff f_n(T) \text{ is irreducible in } \mathbb{Q}[T].}$$

The reverse direction is immediate from

$$f_n = \prod_p f_p^{v_p(n)}.$$

Distinct primes give distinct atomic front polynomials because evaluation at 2 satisfies

$$f_p(2) = p.$$

Therefore the numerical Pratt RRFS is front-faithful.

6.2 Zeta preservation

The same evaluation theorem gives

$$\boxed{f_n(2) = n \quad \text{for every } n \geq 1.}$$

Thus the ordinary norm $N(n) = n$ is evaluation-compatible at $u = 2$. Theorem 5.3 yields

$$\boxed{\sum_{n \geq 1} n^{-s} = \sum_{f_n} f_n(2)^{-s} = \zeta(s).}$$

Hence the natural-number Pratt RRFS and its front-polynomial lift have the same transition matrix, the same recursive Green geometry, the same atomic factorization pattern, and the same analytic zeta function.

This gives a structural interpretation of the f_n family: the irreducibility theorem is exactly the statement that the terminal-front transformation does not split prime atoms.

7 Further front-stable and front-splitting RRFS

Front stability is not restricted to the numerical Pratt system.

7.1 Chains

Let α_0 be terminal and define a chain

$$D\alpha_0 = 1, \qquad D\alpha_j = \alpha_{j-1} \quad (j \geq 1).$$

Then

$$\mathcal{F}_{\alpha_0}(T) = T$$

and recursively

$$\mathcal{F}_{\alpha_j}(T) = 1 + \mathcal{F}_{\alpha_{j-1}}(T) = T + j.$$

Every atomic front polynomial is linear and different from the others. Thus every finite truncation and the infinite chain are front-faithful.

7.2 Cyclotomic stars

Let τ be terminal and let α_r satisfy

$$D\alpha_r = \tau^{2^r}.$$

Then

$$\boxed{\mathcal{F}_{\alpha_r}(T) = 1 + T^{2^r} = \Phi_{2^{r+1}}(T),}$$

a cyclotomic polynomial, hence irreducible. Taking one atom for each exponent 2^r produces another front-faithful family.

More generally, if an atom β has $\mathcal{F}_\beta(T) = T + j$ and

$$D\alpha = \beta^{2^r},$$

then

$$\mathcal{F}_\alpha(T) = 1 + (T + j)^{2^r}$$

is irreducible because translation $T \mapsto T + j$ is an automorphism of $\mathbb{Z}[T]$ and $1 + T^{2^r}$ is irreducible.

7.3 A minimal splitting example

Let τ be terminal and

$$D\alpha = \tau^3.$$

Then

$$\mathcal{F}_\alpha(T) = 1 + T^3 = (T + 1)(T^2 - T + 1).$$

Thus a two-atom finite RRFS can already be front-splitting. Endlichkeit is therefore irrelevant to front stability.

8 Polynomial RRFS over finite fields

We now turn to a natural RRFS whose original atoms are monic irreducible polynomials over a finite field. This should not be confused with the Frobenius Orbit System for finite fields; here the objects are monic polynomials and the recursion comes from polynomial predecessors.

Fix q and let

$$H_q = \mathbb{F}_q[x]_{\text{monic}}$$

with monic irreducible polynomials as atoms. Let x be terminal and define for an irreducible $\phi \neq x$

$$D\phi = \phi - \phi(0).$$

Since $x \mid D\phi$, every irreducible factor of $D\phi$ other than the terminal factor has smaller degree than ϕ; with the terminal convention this gives a finite ranked recursion.

The front polynomial is therefore defined from complete fronts exactly as before and satisfies

$$\mathcal{F}_x(T) = T, \qquad \mathcal{F}_\phi(T) = 1 + \prod_\psi \mathcal{F}_\psi(T)^{v_\psi(D\phi)}.$$

8.1 Failure of faithfulness

If $q > 2$, the distinct linear atoms

$$x - a, \qquad a \in \mathbb{F}_q^\times,$$

all satisfy

$$D(x - a) = x.$$

Hence

$$\boxed{\mathcal{F}_{x-a}(T) = 1 + T}$$

for every $a \neq 0$. Thus the one-variable front transformation already merges distinct atoms and is not front-faithful for $q > 2$.

8.2 Exact front splitting over $\mathbb{F}_3$

Consider

$$\phi(x) = x^3 + x^2 - x + 1 \in \mathbb{F}_3[x].$$

This polynomial is irreducible over $\mathbb{F}_3$. Its predecessor factors as

$$D\phi = x(x^2 + x - 1).$$

Set

$$\psi = x^2 + x - 1.$$

Then ψ is irreducible and

$$D\psi = x(x + 1).$$

Consequently

$$\mathcal{F}_x = T, \qquad \mathcal{F}_{x+1} = 1 + T,$$

$$\mathcal{F}_\psi = 1 + T(1+T) = T^2 + T + 1,$$

and finally

$$\mathcal{F}_\phi = 1 + T(T^2 + T + 1) = T^3 + T^2 + T + 1.$$

But

$$\boxed{T^3 + T^2 + T + 1 = (T+1)(T^2+1).}$$

Thus an irreducible polynomial atom can have a reducible terminal-front polynomial. The finite-field polynomial RRFS is therefore front-splitting in general.

A second example over $\mathbb{F}_3$ is the irreducible polynomial

$$x^4 + x^2 - 1,$$

whose front polynomial is

$$T^4 + T^2 + 1 = (T^2 - T + 1)(T^2 + T + 1).$$

8.3 The classical zeta and the front-evaluation zeta differ

The standard multiplicative norm is

$$N(f) = q^{\deg f},$$

which yields

$$\sum_{f \text{ monic}} q^{-s \deg f} = \frac{1}{1 - q^{1-s}}.$$

Could this norm arise from a single front evaluation $\mathcal{F}_f(u)$? The terminal atom x forces $u = q$, because $\mathcal{F}_x(u) = u = N(x) = q$. But for any nonzero linear atom $x - a$,

$$\mathcal{F}_{x-a}(q) = q + 1 \neq q = N(x-a).$$

Hence the degree norm is not evaluation-compatible with the terminal-front polynomial. The classical finite-field zeta and the natural front-evaluation zeta are therefore different constructions.

9 Dedekind ideals and ideal-Pratt front polynomials

Let $K/\mathbb{Q}$ be a number field and $\mathcal{O}_K$ its ring of integers. The monoid H_K of nonzero integral ideals is factorial:

$$\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{v_\mathfrak{p}(\mathfrak{a})}.$$

Thus nonzero prime ideals are natural RRFS atoms.

For a prime ideal $\mathfrak{p}$ above a rational prime p, define the ideal-Pratt predecessor

$$\boxed{D_K(\mathfrak{p}) = (p-1)\mathcal{O}_K.}$$

Unique factorization of ideals determines its child multiplicities. Every prime ideal below a rational prime $q \mid p - 1$ has underlying rational prime $q < p$, so the recursion is ranked by the numerical Pratt order or by any compatible rank on the underlying rational prime.

The terminal atoms are the prime ideals above 2, because $(2-1)\mathcal{O}_K = \mathcal{O}_K$. The one-variable front polynomial counts all selected terminal prime ideals with the same variable T.

9.1 Gaussian integers

Take

$$K = \mathbb{Q}(i), \qquad \mathcal{O}_K = \mathbb{Z}[i],$$

and let

$$\mathfrak{t} = (1+i).$$

Since $(2) = \mathfrak{t}^2$, $\mathfrak{t}$ is the unique prime ideal above 2 and is terminal. Hence

$$\mathcal{F}_{\mathfrak{t}}(T) = T.$$

The rational prime 3 is inert, so

$$\mathfrak{q} = (3)$$

is a prime ideal and

$$D\mathfrak{q} = (2) = \mathfrak{t}^2.$$

Therefore

$$\boxed{\mathcal{F}_{\mathfrak{q}}(T) = 1 + T^2,}$$

which is irreducible.

The prime 5 splits. For either prime ideal $\mathfrak{p}_{\pm}$ above 5,

$$D\mathfrak{p}_{\pm} = (4) = \mathfrak{t}^4,$$

so

$$\boxed{\mathcal{F}_{\mathfrak{p}_{\pm}}(T) = 1 + T^4.}$$

These polynomials are irreducible, but they are equal. Thus the one-variable front map is already non-faithful: it forgets which conjugate prime ideal lies at the root.

Now consider the inert prime 7 and its prime ideal

$$\mathfrak{r} = (7).$$

Since

$$D\mathfrak{r} = (6) = (2)(3) = \mathfrak{t}^2\mathfrak{q},$$

we obtain

$$\begin{aligned}\mathcal{F}_{\mathfrak{r}}(T) &= 1 + \mathcal{F}_{\mathfrak{t}}(T)^2\mathcal{F}_{\mathfrak{q}}(T)\\ &= 1 + T^2(1+T^2)\\ &= T^4 + T^2 + 1.\end{aligned}$$

But

$$\boxed{T^4 + T^2 + 1 = (T^2 + T + 1)(T^2 - T + 1).}$$

Thus a genuine prime ideal can be front-splitting.

9.2 Comparison with the Dedekind zeta norm

The classical norm is

$$N\mathfrak{a} = |\mathcal{O}_K/\mathfrak{a}|,$$

and the corresponding zeta function is

$$\zeta_K(s) = \sum_{\mathfrak{a}\neq 0}(N\mathfrak{a})^{-s} = \prod_{\mathfrak{p}}(1-(N\mathfrak{p})^{-s})^{-1}.$$

For $\mathbb{Z}[i]$, evaluation compatibility with the front polynomial already fails at the first inert prime. Since $\mathcal{F}_{\mathfrak{t}}(T) = T$ and $N\mathfrak{t} = 2$, a compatible single evaluation would have to use $u = 2$. But

$$\mathcal{F}_{(3)}(2) = 1 + 2^2 = 5, \qquad N(3) = 9.$$

Hence the Dedekind norm is not the front-evaluation norm. The ideal-Pratt front zeta is therefore a new distribution rather than another presentation of $\zeta_{\mathbb{Q}(i)}(s)$.

This contrasts sharply with the numerical Pratt system, where evaluation at 2 simultaneously distinguishes the atomic front polynomials and exactly recovers the original norm.

10 Comparison

The examples can be summarized as follows.

RRFS	Atomic front behavior	Faithful?	Natural zeta transfer?
Natural numbers	$f_p(T)$ irreducible for every prime p	Yes; $f_p(2) = p$ separates atoms	Yes, at $T = 2$: $\zeta(s)$
Chain systems	$T + j$	Yes	For any chosen compatible evaluation gauge
Cyclotomic stars	$1 + T^{2^r}$ irreducible	Yes if the exponents are distinct	Depends on the chosen norm
$\mathbb{F}_q[x]$ polynomial RRFS	Splitting occurs; e.g. over $\mathbb{F}_3$, T^3+T^2+T+1	No for $q > 2$ already at linear atoms	No for the degree norm
Ideal-Pratt over $\mathbb{Z}[i]$	Splitting occurs at (7)	No: conjugate primes over 5 have the same front polynomial	No for the ideal norm

Two distinct obstructions should be kept separate.

1. **Splitting:** one original atom has a reducible front polynomial. The compressed front statistic creates more algebraic irreducible factors than original atoms.
2. **Collision:** two distinct original atoms have the same irreducible front polynomial. The compressed statistic loses root-label information.

Front faithfulness requires that neither obstruction occur. Zeta transfer requires one more ingredient: the original norm must also be recovered by a natural evaluation of the front polynomial.

11 Open problems

Problem 11.1 (Classification of front-stable trees)**.** *Give structural conditions on the recursive rooted tree T_α that imply irreducibility of $\mathcal{F}_\alpha(T)$. In particular, classify the tree shapes for which the terminal-front polynomial is irreducible over $\mathbb{Z}$.*

Problem 11.2 (Front stability for numerical Pratt trees)**.** *Jonathan Love's theorem proves front stability for every numerical prime atom. Find a proof formulated directly in RRFS/front language and isolate which special features of the numerical Pratt recursion force irreducibility.*

Problem 11.3 (Finite-field stability densities)**.** *For fixed q and degree d, let*

$$\eta_q(d) = \frac{\#\{\phi \text{ monic irreducible} : \deg\phi = d,\ \mathcal{F}_\phi \text{ irreducible}\}}{\#\{\phi \text{ monic irreducible} : \deg\phi = d\}}.$$

Study $\eta_q(d)$ as $d \to \infty$ and determine how it depends on q.

Problem 11.4 (Ideal-Pratt splitting)**.** *For a number field K, characterize the prime ideals $\mathfrak{p}$ for which the ideal-Pratt front polynomial $\mathcal{F}_\mathfrak{p}(T)$ is irreducible. Relate the splitting defect $\delta_K(\mathfrak{p})$ to ramification and splitting of the rational primes appearing in the Pratt tree below $p = \mathfrak{p} \cap \mathbb{Z}$.*

Problem 11.5 (Front zeta functions)**.** *For $u > 1$, study*

$$\zeta_{\mathrm{fr},\mathfrak{R}}(s;u) = \prod_{\alpha\in\mathcal{A}} \left(1 - \mathcal{F}_\alpha(u)^{-s}\right)^{-1}.$$

Determine its abscissa of convergence, analytic continuation, prime number laws, and dependence on the recursive geometry. Identify RRFS for which a canonical value of u recovers a classical zeta or L-function.

Problem 11.6 (Iterated front transforms)**.** *For front-faithful RRFS the front lift has the same transition matrix as the original system. Study iteration of the front construction, possible fixed points, and equivalence classes under repeated front lifts.*

12 Conclusion

The front polynomial of an RRFS is most naturally defined by geometry rather than recursion. One first forms the finite recursive forest, then considers its complete node fronts, and finally records the number of selected terminal vertices:

$$\mathcal{F}_h(T) = \sum_{A\in\mathrm{CF}(F_h)} T^{\lambda(A)}.$$

The familiar recursion

$$\mathcal{F}_\tau = T, \qquad \mathcal{F}_\alpha = 1 + \prod_\beta \mathcal{F}_\beta^{B_{\beta,\alpha}}, \qquad \mathcal{F}_h = \prod_\alpha \mathcal{F}_\alpha^{v_\alpha(h)}$$

is then a theorem expressing the two ways a complete front can meet an atomic tree: at the root or independently inside every child subtree.

This distinction clarifies the numerical Pratt polynomials. Their recursion and their node-front interpretation are two descriptions of the same object. Jonathan Love's irreducibility theorem says that this geometric front transform preserves the prime atoms, and the identity $f_n(2) = n$ says that it simultaneously preserves the natural norm. Consequently the natural-number RRFS and its polynomial front lift have the same factorization data and the same Riemann zeta function.

Other RRFS behave differently. Chains and cyclotomic stars give broad front-stable families, while finite-field polynomial systems and Dedekind ideal-Pratt systems show genuine front splitting and atom collisions. Thus front stability is neither automatic nor a consequence of finiteness. It is an arithmetic property of the recursive geometry after terminal-label compression.

The resulting research program is concrete: classify front-stable RRFS, measure front splitting, determine when the front lift is faithful, and understand when a natural front evaluation transfers a classical zeta function. In this sense the front polynomial provides a new bridge from recursive trees to ordinary polynomial factorization and analytic prime distributions.

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